KERAAL team roles: Uijin (data) · Roshini (programming) · Joshi

Next steps after the confirmatory analysis · as of 12 Oct 2026

Where we are. The pipeline and the pre-registered confirmatory analysis are done. H1-H3 were not supported; the likely reason is that **the decision threshold learned from healthy actors (Group 3) does not fit patients**. The next goal is to make this threshold story solid with data. Uijin prepares and links the data; Roshini builds the experiments and statistical models on it.

Member	Role	This week (by Fri 16 Oct)	Next week (by 23 Oct)
Uijin	Data	Cleaned table v2, temporal-segment table	Vicon-Kinect alignment, 2B reference
Roshini	Programming	Threshold-calibration experiment, P1 raw check	Annotator-threshold mixed model, temporal model
Joshi (first-year)	Observation, literature	B1-B3 video check, email draft (separate sheet KERAAL_FirstYear_Tasks)	-

Common rules

- Every new experiment gets a **new config file** in configs/. `configs/p1_confirmatory.json` is locked by the pre-registration; editing it fails the unit tests.
- Everything after the confirmatory run is **exploratory**; label results that way.
- New scripts get a `--check` self-test and are added to src/verify.py. `python src/verify.py` must pass before sharing.
- Always split by participant; report majority baseline, permutation-null 95th percentile and effective N next to any score; report both annotators.
- Log what you did and found in PROGRESS.md with a date; see CODE_GUIDE.md for the code.
- Never upload raw data (CC BY-NC-SA) anywhere, including raw/ and data/ to GitHub.

Uijin: data

Make the data trustworthy and hand Roshini the tables the analyses need.

U1. Cleaned dataset v2 (input_data)

Priority	High	Target	Wed 14 Oct

- **Why**: The 8 Oct review found values and labels correct, but four things need fixing.
- **Steps**: (1) split 2A S1 by site (2A_S1_Brest, 2A_S1_Roscoff) until identity is confirmed; (2) add annotator-B recording-level labels (for A/B analyses); (3) add a training-filter column: drop Incomplete/Motionless/Other, keep in_annotated_span=yes; (4) correct the note saying there is no Temporal-error track (there are 98 segments) and list the two annotator-B files with corrupted end times
- **Deliverable**: model_input_kinect_v2.csv, model_input_blazepose_v2.csv, change note
- **Done when**: v2 Kinect values match the originals (rerun the existing check); participant keys follow data/master_index.csv

U2. Temporal-segment table (15 fps)

Priority	High	Target	Fri 16 Oct

- **Why**: Annotator A's error segments cover a median 40% of a recording; whole-recording summaries dilute the signal, so segment-level analysis needs a clean table.
- **Steps**: one row per segment for 1A/2A: start/end in seconds and frames (seconds × 15), error type, body part, severity, from both Global-error and Temporal-error tracks; check segments outside the annotated span and overlaps
- **Deliverable**: data/temporal_segments.csv plus a QC summary
- **Done when**: all 98 Temporal-error segments present, frame ranges within each recording

U3. Group 3 Vicon-Kinect alignment

Priority	Medium	Target	Fri 23 Oct

- **Why**: To show that the 12° expert-threshold gap on RTK exceeds Kinect measurement error we need the motion-capture (Vicon) comparison; it directly supports the paper's main claim.
- **Steps**: align E:/KERAAL_data/extracted/group3/vicon (540 files, ~2× the Kinect frame count) with Kinect in time, and map the 17 Vicon markers for trunk, shoulders and arms to Kinect joints
- **Deliverable**: list of aligned recording pairs, marker-to-joint map, method note
- **Done when**: all 540 pairs aligned; Roshini can use the map directly

U4. Group 2B reference distribution

Priority	Medium	Target	Fri 23 Oct

- **Why**: 2B are healthy people in the robot-coached sessions (no labels); they separate the session effect from the patient effect between Group 3 and 1A.
- **Steps**: bring the 149 2B Kinect files into the pipeline format and compare camera geometry, torso length and the key features (max shoulder elevation, trunk-rotation range, lateral-tilt range) across groups
- **Deliverable**: per-group distribution table (G3, 2B, 2A, 1A)
- **Done when**: all 149 recordings included, same format as explore/camera_geometry.py

Roshini: programming

Test the threshold hypothesis with experiments and statistical models, through new configs for run_experiment.py or new scripts.

R1. Threshold-calibration experiment (backlog 28)

Priority	High	Target	Fri 16 Oct

- **Why**: In the confirmatory run the Group 3 model labelled 92-97% of test recordings as errors. Whether re-fitting only the decision threshold on real data restores performance is the most direct test of the hypothesis.
- **Steps**: freeze the Group 3 models (CTK C vs E13, R1 and Rerr, RF and logistic regression); for each test participant, refit only the probability threshold on the other test participants and evaluate the held-out one; compare before/after with the participant permutation null and paired bootstrap
- **Deliverable**: src/threshold_calibration.py, a new config, result table
- **Done when**: --check passes, pre-calibration numbers reproduce the confirmatory results, results logged in PROGRESS.md as exploratory

R2. Why raw coordinates did well in P1 (backlog 27)

Priority	High	Target	Fri 16 Oct

- **Why**: R0 (raw) with RF scored 0.737 in P1 but 0.545 with logistic regression; without a confound check the result cannot be interpreted.
- **Steps**: apply person_confound.py's static feature sets to P1 and test R0 against confirmatory.py's fixed-prediction permutation null
- **Deliverable**: P1 confound table
- **Done when**: P1 scores for each static set alongside the null 95th percentile

R3. Annotator-threshold mixed model

Priority	Medium	Target	Fri 23 Oct

- **Why**: The key finding (a 12° threshold gap between annotators on RTK) needs a formal model for the paper; the current logistic fit plus bootstrap is unstable when there are few participants (ELK).
- **Steps**: a logistic mixed model of label ~ key feature + annotator-specific threshold + (1 | participant) (statsmodels or R lme4/ordinal); also split the feature into a person mean and a within-person deviation to see which one drives the label (capacity vs mistake)
- **Deliverable**: explore/annotator_thresholds_mixed.py and a results table (3 exercises × 2 annotators)
- **Done when**: RTK result compared with the bootstrap estimate (+12.0° [+5.8, +15.6]) and logged

R4. Temporal localisation model (after Uijin's U2)

Priority	Medium	Target	Fri 23 Oct

- **Why**: Errors occupy only part of a recording; locating when they occur gives concrete feedback and avoids diluting the signal.
- **Steps**: start simple: predict per-second error from Uijin's segment table and cleaned v2 using summary statistics + RF/logistic regression; participant-level splits; per-second Macro-F1 and a segment-overlap metric
- **Deliverable**: src/temporal_baseline.py and a results table
- **Done when**: includes a no-participant-overlap check; --check passes

Hand-offs

When	From → to	What
Wed 14 Oct	Uijin → Roshini	Cleaned table v2 (U1)
Fri 16 Oct	Uijin → Roshini	Temporal-segment table (U2) → R4 starts
Fri 16 Oct	Roshini → team	Calibration and P1 confound results (R1, R2)
Fri 23 Oct	Uijin → Roshini	Vicon alignment and 2B table (U3, U4) → measurement-error analysis
Fri 23 Oct	Roshini → team	Mixed-model and temporal results (R3, R4)

References: PROGRESS.md (log), PREREGISTRATION.md (confirmatory results), RESEARCH_IDEAS.md (directions), CODE_GUIDE.md (code), KERAAL_Team_Brief_KR_EN.pdf (team brief).